

Convolution Neural Networks

Applications: transfer learning, face recognition, neural-style transfer

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Transfer learning: Problem statement

Quick remark

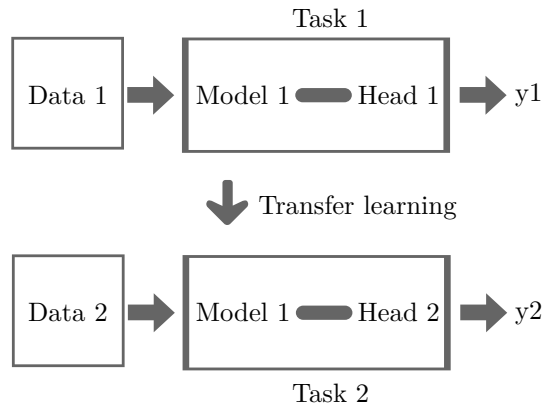
- You should see code demonstration uploaded on Moodle some time ago.
- We use the example presented in the uploaded Jupyter Notebook.

Problem statement

- We are given a very small dataset of bees and ants.
- Train a neural network (convolution neural network) to perform the classification task on this small dataset

Key issue *very small dataset*

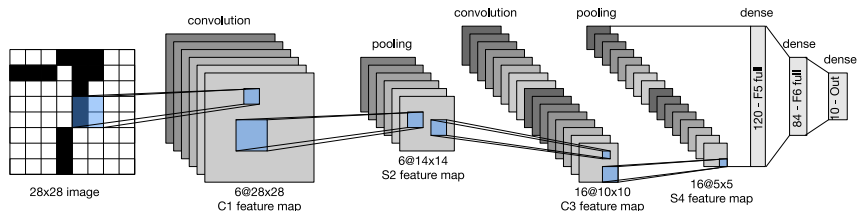
Transfer learning: General idea



Task 1 – Model 1 + Head 1: classify many objects (perhaps ≥ 1000 objects)

↓
Task 2 – Model 1 + Head 2: classify just bees and ants

Transfer learning: Model & Head

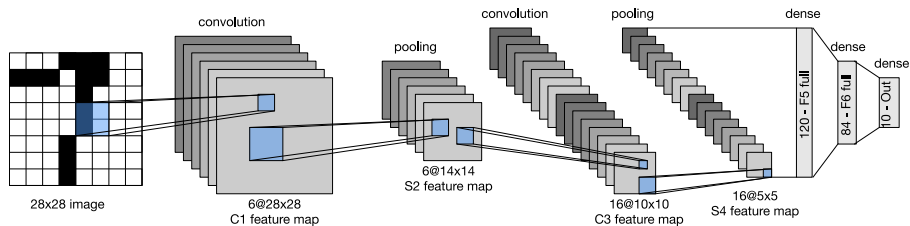


- In general a model refers to everything in the neural network.
- For now, we refer to convolution layers, pooling layers and several layers before the fully connected layer as '**model**'.
- The last or the few last fully connected layers as '**head**'.

Key point

- You want to change a small part, and keep the rest, of a very deep convolution neural network to fit your problem statement.
- The 'head' and the 'main body' can be defined by the practitioner in a custom fashion.

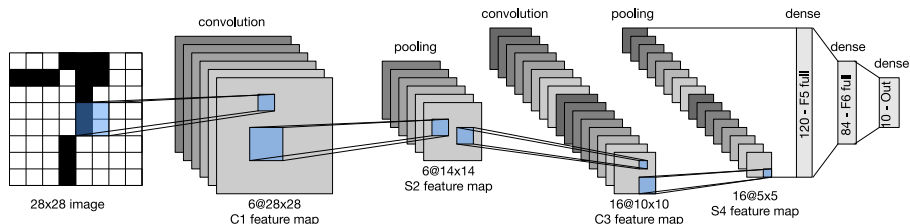
Transfer learning: Model & Head



What would we do in our “bee-vs-ant” classification problem?

Various ways – (Reminder for myself: Lecturer must draw on this slide):

Transfer learning: Model & Head

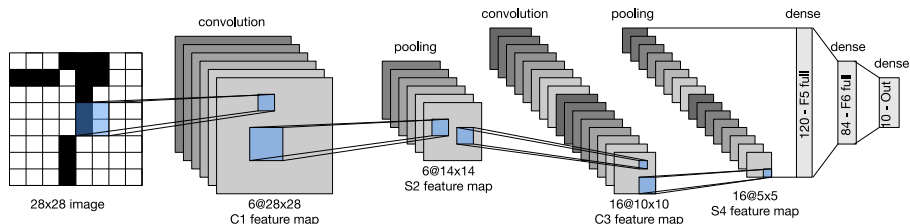


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- Replace the last fully connected layer by a fully connected layer with only 1 output (probability that $y = 1$).

Transfer learning: Model & Head

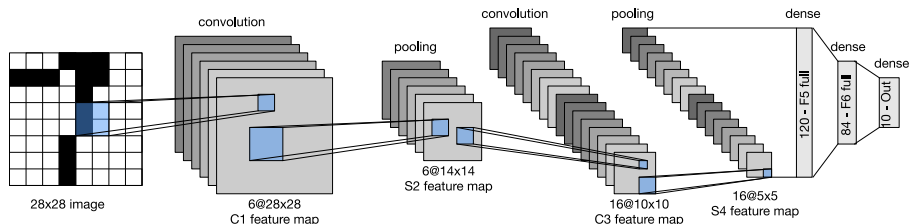


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- Replace the last fully connected layer by a fully connected layer with only 1 output (probability that $y = 1$).
- Replace the last fully connected layer and even two fully connected layers before it using different numbers of input and output features.

Transfer learning: Model & Head



What would we do in our “bee-vs-ant” classification problem?

Various ways – (Reminder for myself: Lecturer must draw on this slide):

- Replace the last fully connected layer by a fully connected layer with only 1 output (probability that $y = 1$).
- Replace the last fully connected layer and even two fully connected layers before it using different numbers of input and output features.
- Replace and add more layers if we want.

Data augmentation

❑ What is data augmentation?

Oxford dictionary

augmentation

noun [C or U]

UK  /ˌɔːɡ.menˈteɪ.jən/ US  /ˌɑːɡ.menˈteɪ.jən/

Add to word list 

the process of increasing the size, value, or quality of something by adding to it:

- *I think of making computers smarter as intelligence augmentation.*
- *Cosmetic procedures include nose reshaping, breast augmentation, and face-lift surgery.*

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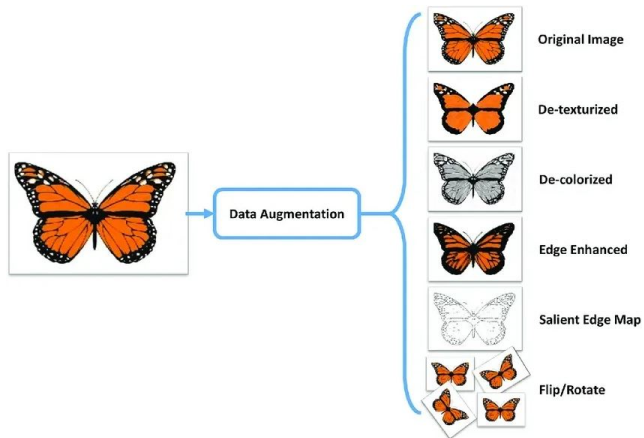
the process of increasing the size, value, or quality of something by adding to it:

- *I think of making computers smarter as intelligence augmentation.*
- *Cosmetic procedures include nose reshaping, breast augmentation, and face-lift surgery.*

❑ Why do we want data augmentation?

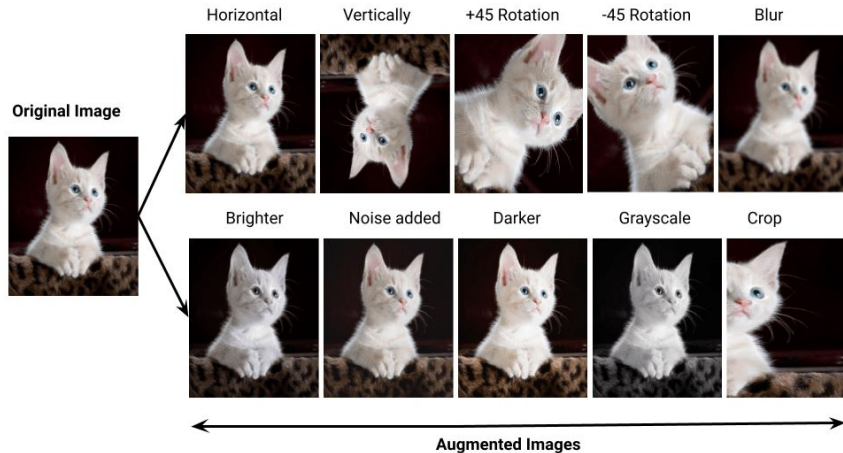
- We have a small dataset of images.
- We want to generate more data from the original set of images.
- The more data the model is trained on, the better the generalization the model can reach.

Data augmentation: Quick examples



Data augmentation: Butterfly example

Data augmentation: Quick examples



Data augmentation: Cat example

Data augmentation in PyTorch

Getting started with transforms v2

• NOTE

Try on [Colab](#) or [go to the end](#) to download the full example code.

This example illustrates all of what you need to know to get started with the new `torchvision.transforms.v2` API. We'll cover simple tasks like image classification, and more advanced ones like object detection / segmentation.

First, a bit of setup

```
from pathlib import Path
import torch
import matplotlib.pyplot as plt
plt.rcParams["savefig.bbox"] = 'tight'

from torchvision.transforms import v2
from torchvision.io import decode_image

torch.manual_seed(1)

# If you're trying to run that on Colab, you can download the assets and the
# helpers from https://github.com/pytorch/vision/tree/main/gallery/
from helpers import plot
img = decode_image(str(Path('../assets') / 'astronaut.jpg'))
print(f"type(img) = {type(img)}, dtype = {img.dtype}, shape = {img.shape}")
```

Note that the most updated version is transformation v2. [Click on Me!](#)

Data augmentation in PyTorch

Illustration of transforms

• NOTE

Try on [Colab](#) or [go to the end](#) to download the full example code.

This example illustrates some of the various transforms available in [the torchvision.transforms.v2 module](#).

```
from PIL import Image
from pathlib import Path
import matplotlib.pyplot as plt

import torch
from torchvision.transforms import v2

plt.rcParams["savefig.bbox"] = 'tight'

# if you change the seed, make sure that the randomly-applied transforms
# properly show that the image can be both transformed and *not* transformed!
torch.manual_seed(0)

# If you're trying to run that on Colab, you can download the assets and the
# helpers from https://github.com/pytorch/vision/tree/main/gallery/
from helpers import plot
orig_img = Image.open(Path('../assets') / 'astronaut.jpg')
```

Personally, I'd like to start from here to learn it quickly. [Click on Me!](#)

Data augmentation in PyTorch

⇒ *Key messages:*

- Too many transformations to learn.
- Only need to understand the mechanisms.
- Learn a few useful and frequently transformations.
- Look up transformations when needed before writing your own code to do that.

⇒ *Some useful transformations* (bullets contain links to PyTorch documentation, just click on them!)

- ☆ [Normalize](#)
- ☆ [Random Crop](#)
- ☆ [Random Resized Crop](#)
- ☆ [Random Horizontal Flip](#)
- ☆ [Random Vertical Flip](#)

Transformation examples using `torchvision.transforms.v2`



Original image

Transformation examples using torchvision.transforms.v2

⇒ Python code for making transformation using torchvision.transforms.v2

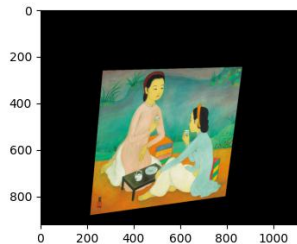
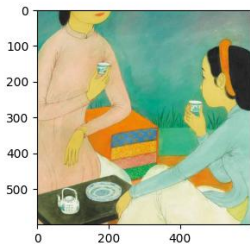
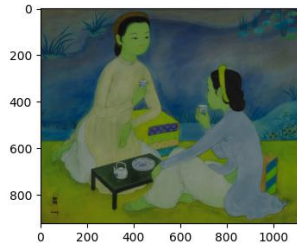
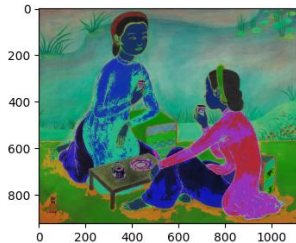
```
from torchvision.transforms import v2
file_path = './figures/painting-vietnamese-ladies.jpg'
original_image = Image.open(file_path)

transformer_list = [v2.RandomSolarize(threshold=192.0),
                    v2.ColorJitter(brightness=.5, hue=.3),
                    v2.CenterCrop(size=600),
                    v2.RandomPerspective(distortion_scale=0.6, p=1.0)]

plt.figure(figsize=(8, 6))
for j in range(len(transformer_list)):
    plt.subplot(2, 2, j+1)
    transformed_image = transformer_list[j](original_image)
    plt.imshow(transformed_image)
plt.tight_layout()
```

⇒ For more fun, please visit [Illustrations of transforms](#) and of course our Moodle.

Transformation examples using `torchvision.transforms.v2`



Random Solarized (top left) – Color Jitter (top right)
Center Crop (bottom left) – Random Perspective (bottom right)

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Face Verification versus Face Recognition

⇒ Face Verification

- ☆ Input: Image, name/ID
- ☆ Output: Whether the input image is that of the claimed person

⇒ Face Recognition

- ☆ Has a database of K people
- ☆ Get an input image
- ☆ Output: whether the image is any of the K persons (or “*not recognized*”)

One-shot learning

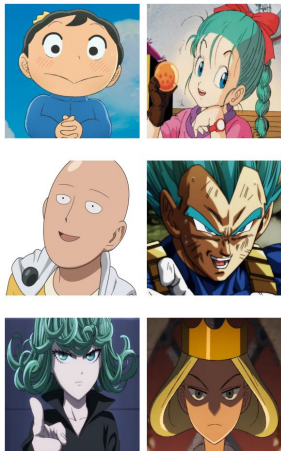


Figure: Database



Yes: Vegeta



No: A clown?

Learning from one example to recognize the person again

Learning a “similiartiy” function

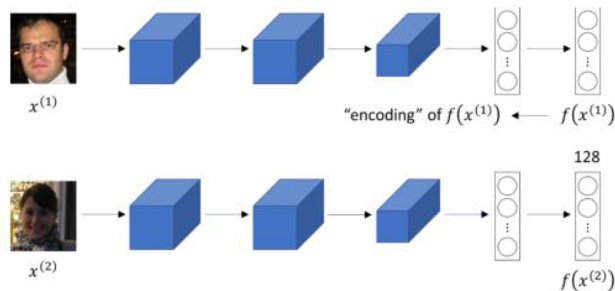
$d(\text{image}(1), \text{image}(2)) = \text{degree of difference between two images}$

$$\begin{cases} \text{if } d(\text{image}(1), \text{image}(2)) \leq \tau & \rightarrow \text{ same} \\ \text{if } d(\text{image}(1), \text{image}(2)) > \tau & \rightarrow \text{ different} \end{cases}$$



Figure: Database

Siamese Network



Reference 1¹ — Reference 2²

⇔ Distance function

$$d(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}) = \|\mathbf{f}(\mathbf{x}^{(1)}) - \mathbf{f}(\mathbf{x}^{(2)})\|^2 = \sum_{i=1}^{n_f} \left[f_i(\mathbf{x}^{(1)}) - f_i(\mathbf{x}^{(2)}) \right]^2$$

¹Learning a similarity metric discriminatively, with application to face verification

²DeepFace: Closing the Gap to Human-Level Performance in Face Verification

Goal of learning

- ⇒ Parameters of neural network define the encoding $\mathbf{f}(\mathbf{x}^{(i)})$
- ⇒ Learn parameters so that:
 - If $\mathbf{x}^{(i)}$ and $\mathbf{x}^{(j)}$ are the same person (same identity),
then $\|d(\mathbf{f}(\mathbf{x}^{(i)}), \mathbf{f}(\mathbf{x}^{(j)}))\|^2$ is **small**.
 - If $\mathbf{x}^{(i)}$ and $\mathbf{x}^{(j)}$ are different persons (different identities),
then $\|d(\mathbf{f}(\mathbf{x}^{(i)}), \mathbf{f}(\mathbf{x}^{(j)}))\|^2$ is **large**.

Learning Objective



(a) Anchor (A)



(b) Positive (P)



(a) Anchor (A)



(b) Negative (N)

⇒ One training example has the form (A, P, N) .

Learning Objective



(a) Anchor (A)



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(a) Anchor (A)



(b) Negative (N)

⇒ One training example has the form (A, P, N) .

⇒ We want distance function satisfying

$$\underbrace{\|\mathbf{f}(A) - \mathbf{f}(P)\|^2}_{d(A,P)} \leq \underbrace{\|\mathbf{f}(A) - \mathbf{f}(N)\|^2}_{d(A,N)} \text{ easily satisfied by setting } d(A, P) = d(A, N)$$

Learning Objective



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⇒ We update the condition with a positive margin $\alpha > 0$

$$\|\mathbf{f}(A) - \mathbf{f}(P)\|^2 - \|\mathbf{f}(A) - \mathbf{f}(N)\|^2 + \alpha \leq 0$$

Triplet loss function

⇒ Given 3 images: Anchor (A), Positive (P), Negative (N) put into one tuple (A, P, N)

³FaceNet: A Unified Embedding for Face Recognition and Clustering

Triplet loss function

- ⇒ Given 3 images: Anchor (A), Positive (P), Negative (N) put into one tuple (A, P, N)
- ⇒ Triplet loss function for one training example – [Reference](#)³:

$$\mathcal{L}(A, P, N) = \max (\|\mathbf{f}(A) - \mathbf{f}(P)\|^2 - \|\mathbf{f}(A) - \mathbf{f}(N)\|^2 + \alpha, 0)$$

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- ⇒ Triplet loss function for a batch of training examples

$$\mathcal{L}(A, P, N) = \sum_{i=1}^m \mathcal{L}(A^{(i)}, P^{(i)}, N^{(i)})$$

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- ⇒ Triplet loss function for a batch of training examples

$$\mathcal{L}(A, P, N) = \sum_{i=1}^m \mathcal{L}(A^{(i)}, P^{(i)}, N^{(i)})$$

- ⇒ Training set: For 1k persons, we may need around 5k – 10k pictures to form dataset of (A, P, N) triplets.

³FaceNet: A Unified Embedding for Face Recognition and Clustering

Triplet loss function

⇒ Triplet Loss in PyTorch: [TripletMarginLoss](#) – [Click on Me!](#)

☆ `TripletMarginLoss` receives the encodings $\mathbf{f}(A), \mathbf{f}(P), \mathbf{f}(N)$ (not the images A, P, N themselves) as inputs.

☆ `TripletMarginLoss` allows p -norm (default $p = 2$ means Euclidean norm):

$$\|\mathbf{f}\|_p = \left(\sum_i^{n_f} f_i^p \right)^{1/p}$$

⇒ For shallow convolution neural network, have a look at [Reference](#)⁴.

⁴Learning local feature descriptors with triplets and shallow convolutional neural networks

Choosing the triplets (A, P, N)

- ⇒ During training, if A, P, N are chosen randomly, it may happen that $d(A, P) + \alpha \leq d(A, N)$ is easily satisfied.

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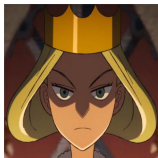
⇒ Choose triplets that are “hard” to train on. Repeat

$$d(A, P) + \alpha \leq d(A, N)$$

⇒ If we choose (A, P, N) so that $d(A, P)$ is quite close to $d(A, N)$, the model will try harder to learn the encodings $\mathbf{f} = \mathbf{f}(\mathbf{x})$ better and better. In other words, the encoding for each person will be more characteristic for that person.

Training set using triplet loss

Anchor



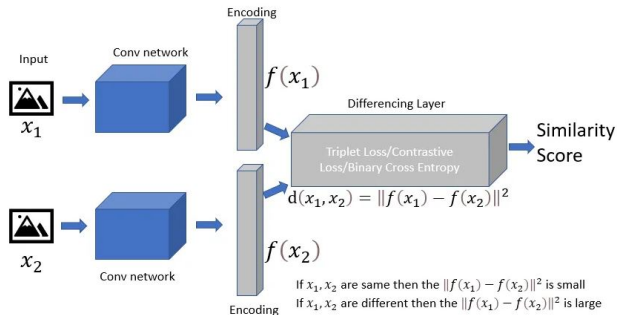
Positive



Negative



Learning the similarity function



A different way to measure similarity score $\rightarrow$ Use binary cross entropy

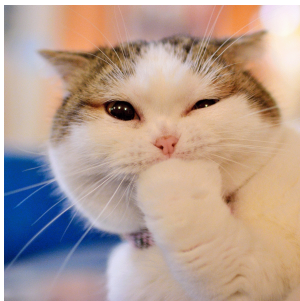
$$\hat{y} = \sigma \left(\sum_{k=1}^{n_f} w_i |f_k(\mathbf{x})^{(i)} - f_k(\mathbf{x}^{(j)})| + b \right)$$

Dataset: pair $\mathbf{X}^{(a)} = (\mathbf{x}^{(i)}, \mathbf{x}^{(j)})$ with label $y^{(a)} = 1$ if $\mathbf{x}^{(i)}$ and $\mathbf{x}^{(j)}$ are the same person,
with label $y^{(a)} = 0$ if $\mathbf{x}^{(i)}$ and $\mathbf{x}^{(j)}$ are two different persons.

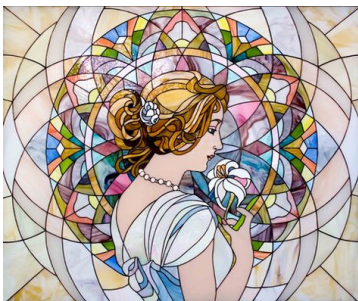
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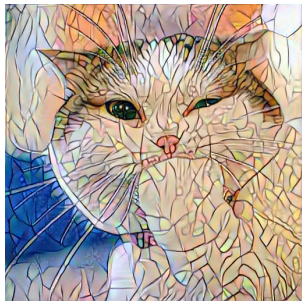
What is neural style transfer



(a) Content Image



(b) Style Image

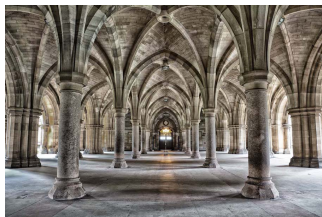


(c) Generated Image

Content Image + Style Image $\longrightarrow$ Generated Image

Reference. [Click on Me!](#)

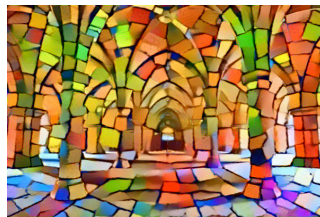
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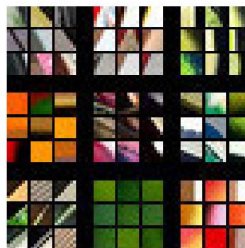
Content Image + Style Image $\longrightarrow$ Generated Image

Reference. Click on Me!

What are deep ConvNets learning?



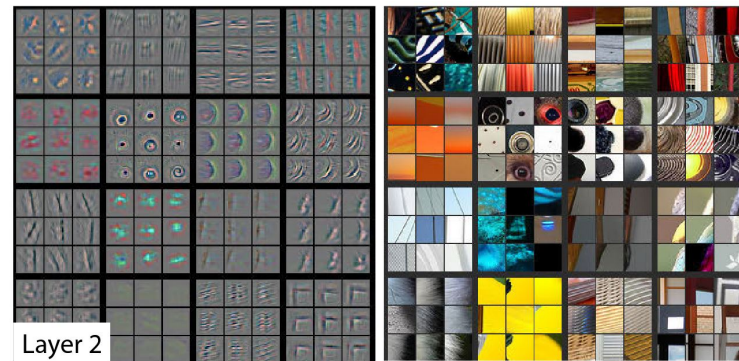
Layer 1



For layers 2-5 the authors showed the top 9 activations in a random subset of feature maps across the validation data, projected down to pixel space using our “deconvolutional” network approach.

Reference. The authors studied the famous AlexNet at the time.

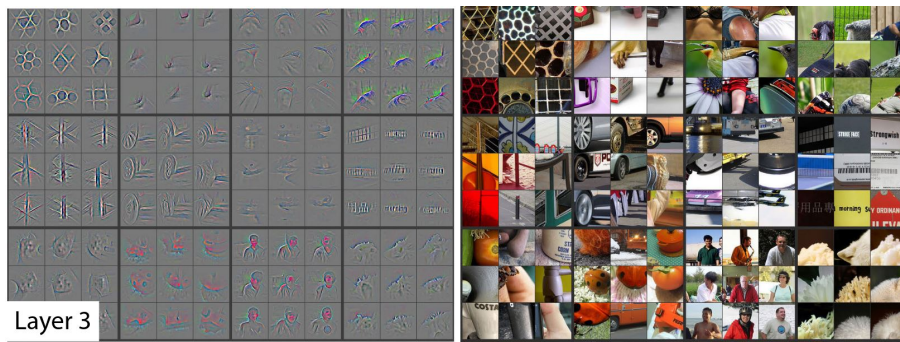
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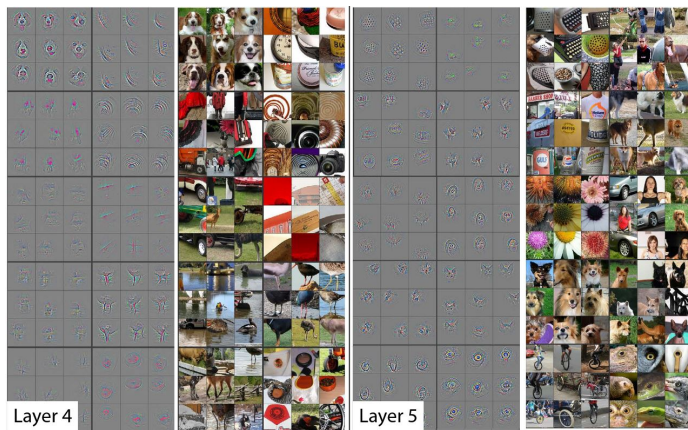
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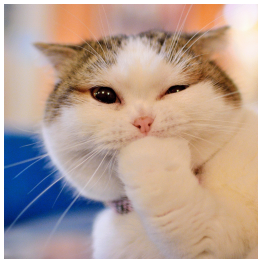
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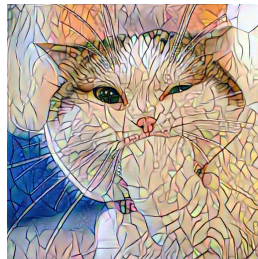
Neural style transfer loss function



(a) Content C

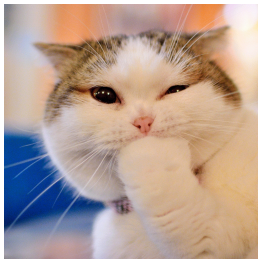


(b) Style S



(c) Generated G

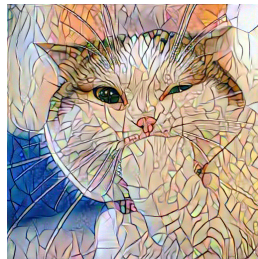
Neural style transfer loss function



(a) Content C



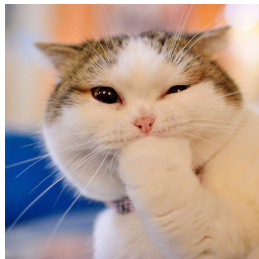
(b) Style S



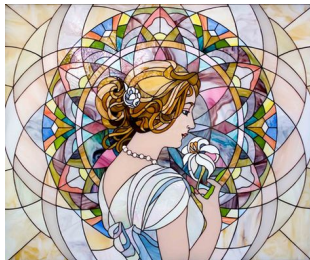
(c) Generated G

$$\mathcal{L}_{\text{content}}(C, G) +$$

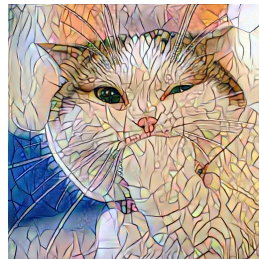
Neural style transfer loss function



(a) Content C



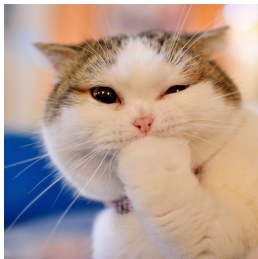
(b) Style S



(c) Generated G

$$\mathcal{L}_{\text{content}}(C, G) + \mathcal{L}_{\text{style}}(S, G)$$

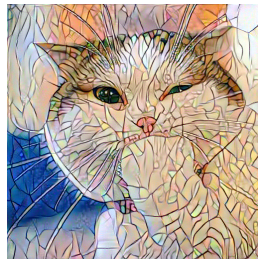
Neural style transfer loss function



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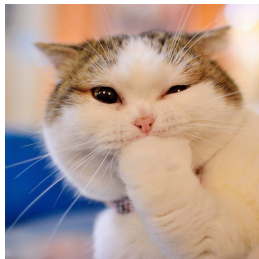
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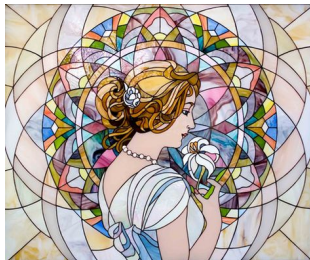
(c) Generated G

$$\mathcal{L}(G) = \mathcal{L}_{\text{content}}(C, G) + \mathcal{L}_{\text{style}}(S, G)$$

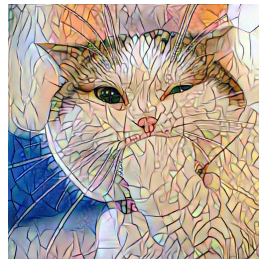
Neural style transfer loss function



(a) Content C



(b) Style S



(c) Generated G

$$\mathcal{L}(G) = \alpha \mathcal{L}_{\text{content}}(C, G) + \beta \mathcal{L}_{\text{style}}(S, G)$$

Find the generated image G

⇨ *Generate the neural-style transfer image G*

① Initialize G randomly

For example: $G : 256 \times 256 \times 3$

② Use optimization method to minimize $\mathcal{L}(G)$

Find the generated image G

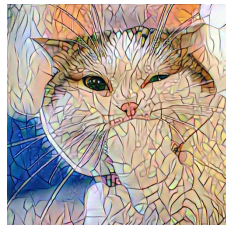
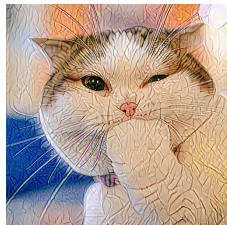
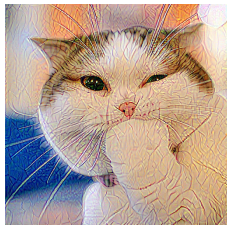
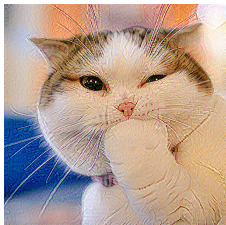
⇒ *Generate the neural-style transfer image G*

① Initialize G randomly

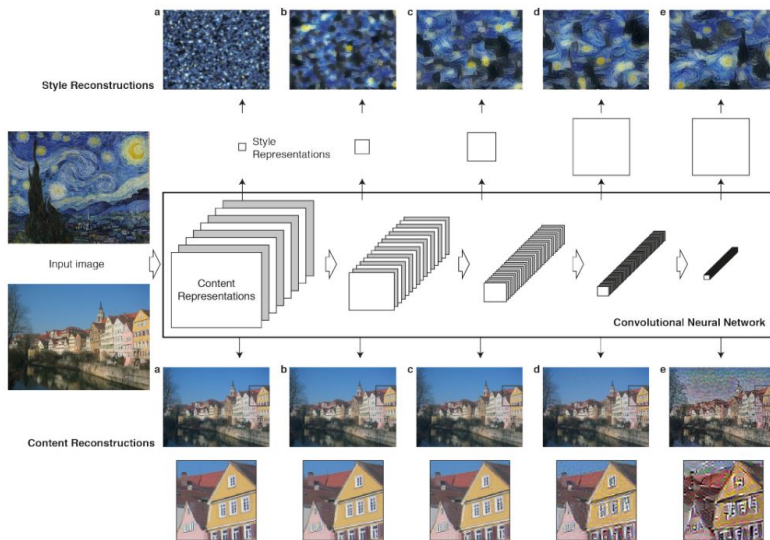
For example: $G : 256 \times 256 \times 3$

② Use optimization method to minimize $\mathcal{L}(G)$

⇒ Example of training process:



Style and Content Reconstruction



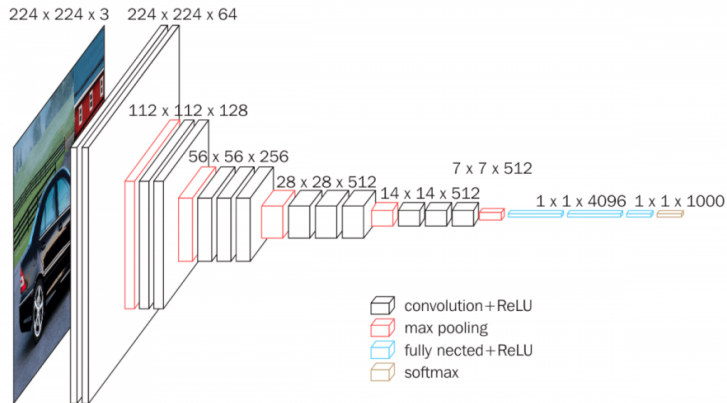
Content loss function

$$\mathcal{L}(G) = \alpha \mathcal{L}_{\text{content}}(C, G) + \beta \mathcal{L}_{\text{style}}(S, G)$$

- Say, e.g., we use hidden layer l to compute content loss
- Use pre-trained ConvNet, .e.g, VGG network ([Reference – VGGs on PyTorch](#))
- Let $\mathbf{a}^{[l](C)}$ and $\mathbf{a}^{[l](G)}$ be the activation of layer l on the images
- If $\mathbf{a}^{[l](C)}$ and $\mathbf{a}^{[l](G)}$ are similar, both images have similar content

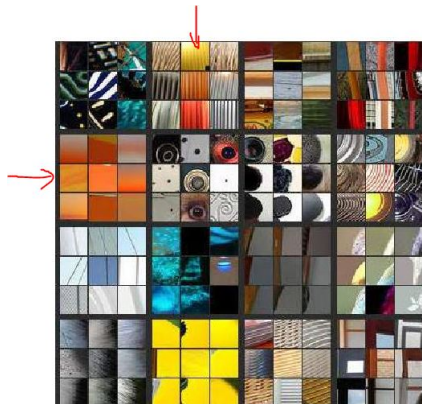
$$\mathcal{L}_{\text{content}}(C, G) = \frac{1}{2} \|\mathbf{a}^{[l](C)} - \mathbf{a}^{[l](G)}\|^2$$

Content loss function



$$\mathcal{L}_{\text{content}}(C, G) = \frac{1}{2} \|\mathbf{a}^{[l](C)} - \mathbf{a}^{[l](G)}\|^2$$

Intuition about style of an image



Are the channels in one layer correlated/uncorrelated?

Are the feature maps in one layer correlated/uncorrelated?

Style matrix

- ⇒ Let $a_{i,j,k}^{[l]}$ denote activation at (i, j, k) in the order (Channel, Height, Width).
Gram matrix:

$$G_{rs}^{[l]} = \sum_{i=1}^{n_H^{[l]}} \sum_{j=1}^{n_W^{[l]}} a_{rij}^{[l]} a_{sij}^{[l]}$$

- ⇒ Gram matrices for style image and generated image:

$$G_{rs}^{[l](S)} = \sum_{i=1}^{n_H^{[l]}} \sum_{j=1}^{n_W^{[l]}} a_{rij}^{[l](S)} a_{sij}^{[l](S)} \qquad G_{rs}^{[l](G)} = \sum_{i=1}^{n_H^{[l]}} \sum_{j=1}^{n_W^{[l]}} a_{rij}^{[l](G)} a_{sij}^{[l](G)}$$

- ⇒ Style loss function for layer $[l]$

$$\mathcal{L}_{\text{style}}^{[l]}(S, G) = \left\| \mathbf{G}^{[l](S)} - \mathbf{G}^{[l](G)} \right\|^2 = \sum_r \sum_s \left(G_{rs}^{[l](S)} - G_{rs}^{[l](G)} \right)^2$$

Style loss function

⇒ Style loss function for layer $[l]$

According to the paper, the authors made normalization:

$$\mathcal{L}_{\text{style}}^{[l]}(S, G) = \frac{1}{2n_H^{[l]}n_W^{[l]}n_C^{[l]}} \sum_r \sum_s (G_{rs}^{[l](S)} - G_{rs}^{[l](G)})^2$$

$$\mathcal{L}_{\text{style}}(S, G) = \sum_l \lambda^{[l]} \mathcal{L}_{\text{style}}^{[l]}(S, G)$$

where $\lambda^{[l]}$ are user-defined weights for computing the style loss (Repeat $[l]$ refers to the l^{th} layer).

⇒ Further notes:

- My implementation does not need those normalization and also weight coefficients $\lambda^{[l]}$.
- These constants pretty much go to the effect of the weights α and β in the total loss function

$$\mathcal{L}^G = \alpha \mathcal{L}_{\text{content}}(C, G) + \beta \mathcal{L}_{\text{style}}(S, G)$$